

1. $s = \text{"Python"}$
0 1 2 3 4 5

`print (s[0], s[-1], s[6])`

* It comes under under error $s[6]$
not in indexing range

2. $s = \text{'Hello'}$
-5 -4 -3 -2 -1

$s[-2] = \text{'l'}$

3. $s = \text{"code"}$

$s[10]$: Index error

$s[1:10]$: ode

4. $s = \text{'mississippi'}$
0 1 2 3 4 5 6 7 8 9 10

first s : $s[2]$

last s : $s[-5]$

5. $S = \text{"a bc def"}$
 4 5 4 3 2 1

`print(S[-1], S[-3], S[-4])`

Ans: f d a

SLICING

BASICS

6. $S[start : step : step]$

↓

It sets the initial here.

7. $S = \text{"Hello World"}$
 0 1 2 3 4 5 6 7 8 9

`print(S[0:5])`

Ans: 'Hello'

`print(S[5:])`

Ans: 'World'

`print(S[:5])`

Ans: 'Hello'

`print(S[:])`

Ans: 'Hello World'

8. `S = "python"`
 0 1 2 3 4 5
`print(S[4:1])`

Ans: b d

9. Last 3 character of string

`print(S[-3])`

10. `S = "a b c d e f g h"`
 0 1 2 3 4 5 6 7

<code>print(S[::1])</code>	<code>print(S[::2])</code>	<code>print(S[1::2])</code>
Ans: 'a b c d e f g h'	Ans: 'a c e g'	Ans: 'b d f h'

STEP & REVERSE SLICING

11. `S = "python"`

`S[::-1]` => 'nohtyp'

12. `S = "a b c d e f g h"`
 0 1 2 3 4 5 6 7
 -8 -7 -6 -5 -4 -3 -2 -1

<code>print(S[::-1])</code>	<code>print(S[::-2])</code>	<code>print(S[6::-1])</code>
Ans: 'h g f e d c b a'	Ans: 'g e c a'	Ans: 'g f e d c b a'

13. `s = "a1b2c3d4e5"`
 0 1 2 3 4 5 6 7 8 9

`S`
`S[1:6]`

`S[1:10:2]`

14. `s = "PYTHON"`
 0 1 2 3 4 5

<code>print(s[5:0:-1])</code>	<code>print(s[5:1-1])</code>	<code>print(s[0:-1])</code>
Ans: 'NOTTYP'	Ans: 'NOTTYP'	Ans: 'NOTTYP'

15. `s = "hello"`
`print(s[100::-1])`
 'olleh'

`print(s[::100])`
 'o'

INTERMEDIATE

16. `s = "abcdefghi"`
 0 1 2 3 4 5 6 7 8 9

`s[2:9]`

Ans: 'cdefghi'

17.

s = "nacear"

s[::-1]

"nacear"

∴ s is a palindrome

18. s = "Hello"

print(s[1:100])

'ello'

print(s[100:200])

' '

print(s[-100:3])

'ello'

19.

s = "HelloWorld"

0 1 2 3 4 5 6 7 8 9

s₂ = s[0:5] + "python"

print(s₂)

"hellopython"

20.

s = "abcdefghij"
0 1 2 3 4 5 6 7 8 9

slice (start: stop: step)

slice (1: 8: 2)

6
"cdgh"